Yang Du

University of Michigan – Ann Arbor – MI, United States

+1 (734)747-3879 • duyung@umich.edu • young-du.github.io

Final year undergraduate at U-M pursuing a major in Computer Science and minor in Mathematics. Research interests: cryptography, security, theoretical computer science, algorithm, computational complexity.

Education

University of Michigan (Ann Arbor) — Michigan, United States
B.S.E. in Computer Science — Major GPA: 4.0/4.0 — *Sep, 2019 - May, 2021*

Shanghai Jiao Tong University — Shanghai, China
B.S.E. in Electrical and Computer Engineering — Major GPA: 3.88/4.0 — *Sep, 2017 - Aug, 2019*

Research Projects

Oblivious Data Structure and Searching Algorithm
Guide: Prof. Elaine Shi, CS Dept, Carnegie Mellon University — *Apr, 2020 - Present*

- Devise an oblivious range searching data structure and a searching algorithm based on path and circuit ORAM, using the idea of k-d tree, range tree, fractional cascading and compare the efficiency. Create a reference implementation.
- Determine a way to obliviously build the oblivious range tree and extend to higher dimension. Find a real life application on Covid-19 exposure tracking.
- Implement an oblivious AVL tree data structure with Path ORAM and reverse deterministic lexicographic eviction.

Efficient Algorithm to find number of prime factors of Integer with Oracle
Guide: Dr. Ilya Volkovich, CSE Dept, University of Michigan — *Apr, 2020 - Sep, 2020*

- Devise the first efficient deterministic algorithm for approximating $\omega(N)$ the number of prime factors of an integer given in addition oracle access to Euler's Totient function.
- Show that the algorithm can be extended to handle a more general class of additive functions that "depend solely on the exponents in the prime factorization of an integer"
- Gives the first algorithm that approximates $\omega(N)$ without necessarily factoring N.

Secure Cloud Manufacturing
Guide: Prof. Z. Morely Mao, EECS Dept, University of Michigan — *Sep, 2020 - Present*

- Perform traffic analysis on PLC network traffic to identify whether the PLC network is normally functioning.
- The traffic analysis results can be used to detect malicious or malfunctioning devices. Train an machine learning classification model on the collected data.
- Compare and figure out which ML algorithm performs best on the traffic data and what kind of features we need to extract from the network traffic.

Free Speech Technology
Guide: Prof. Dick Robert, EECS Dept, University of Michigan — *Dec, 2019 - May, 2020*

- Using Bluetooth technology to broadcast messages, designed to support free speech by preventing blocking, censorship, and surveillance.
- Support communication during events that disable the internet and cellular communication infrastructures.

Publication

Yang Du, Ilya Volkovich, ***Approximating the Number of Prime Factors Given an Oracle to Euler's Totient Function***, ITCS 2021, submitted.

Key Academic Projects

Cryptocurrency and Blockchain
Guide: Prof. Mahdi Cheraghchi, CSE Dept, University of Michigan *Mar, 2020 - Apr, 2020*
- Study Bitcoin (public key encryption) and Z-cash (zero-knowledge proof).
- Blockchain application to e-voting and supply chain tracking.

Deep Learning Techniques for Classification and Feature Learning
Guide: Dr. Kutty Sindhu Krishnan, CSE Dept, University of Michigan *Feb, 2020 - Apr, 2020*
- Use Pytorch to classify pictures different types of food.
- Compare different methods: NN, CNN, Autoencoder.

Teaching Experience and Internship

Linear Algebra, Proof Tutor
Instructor: Dr. Scott Schneider, Maths Dept, University of Michigan *Jan, 2020 - Present*
- Hold office hours to discuss homework questions with students.

Foundation of Computer Science, Grader
Instructor: Dr. Amir Kamil, CSE Dept, University of Michigan *Sep, 2019 - May, 2020*
- Grade homework and exam paper.

Linear Algebra, Teaching Assistant
Instructor: Prof. Olga Danilkina, Shanghai Jiao Tong University *Feb, 2019 - May, 2019*
- Lead weekly discussion session.
- Grade homework and exam paper.

Honors Mathematics II, Teaching Assistant
Instructor: Prof. Horst Hohberger, Shanghai Jiao Tong University *Sep, 2018 - Dec, 2018*
- Lead weekly discussion session.
- Grade homework and exam paper.

Software Engineer of Developing a Teaching Management System
Shanghai Suxun Education Ltd. *Dec, 2018 - Feb, 2019*
- Front end(Android) development.
- Design and format API and design data structure to minimize data transferred.

Scholastic Achievements

- **Roger King Scholarship** awarded by University of Michigan *Aug, 2020*
- **Dean's Honors List** awarded by University of Michigan *Dec, 2019*
- **University Honors** awarded by University of Michigan *Dec, 2019*
- **Fuda Scholarship** awarded by Shanghai Jiao Tong University *Nov, 2018*
- **Bronze Medalist of the 9th University Physics Competition** *Oct, 2018*
- **Undergraduate Merit Scholarship** awarded by Shanghai Jiao Tong University *Nov, 2018 & 2019*

Technical Strengths

- **Programming Languages:** C/C++, Python, Java
- **Development:** Android
- **Others:** LaTeX, Matlab, Mathematica, Origin